💥 CANNON DUEL BATTLE CARD

10 Points Total

My Name: _____ **Rival Partner:** _____ **Period:** _____

1. Open the Game:
rrmudry.github.io/Cannon_duel

2. Pick a Game Mode:
Play **2 Players (1 Device)** with a partner OR battle the **Computer**!

3. Knock Out the Fort:
Use what you learned this week to dial in your angle & blast the other cannon!

Part 1: Your 5-Shot Battle Log (Aim, Fire & Record) — Low Floor • 6 Points

SHOT #	LAUNCH ANGLE (Θ)	LAUNCH SPEED (V_0)	WHAT HAPPENED? (CIRCLE ONE)
Shot 1	_____ °	_____ m/s	Land Short 💥 DIRECT HIT! Fly Too Far
Shot 2	_____ °	_____ m/s	Land Short 💥 DIRECT HIT! Fly Too Far
Shot 3	_____ °	_____ m/s	Land Short 💥 DIRECT HIT! Fly Too Far
Shot 4	_____ °	_____ m/s	Land Short 💥 DIRECT HIT! Fly Too Far
Shot 5	_____ °	_____ m/s	Land Short 💥 DIRECT HIT! Fly Too Far

Part 2: Artillery Physics Strategy (Explain & Sketch) — High Ceiling • 4 Points

QUESTION 1: FIXING A MISS

1. Fixing a Miss: If your cannonball lands **20 meters SHORT** of your target, what is <u>one thing</u> you can change on your next shot to make it reach the fort?

Written Explanation:

✏️ **Draw an Example Sketch:** (Short vs. Fixed Shot)

Sketch trajectory paths:

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QUESTION 2: HIGH ARC VS. FLAT LASER

2. High Arc vs. Flat Laser: Both **35°** and **55°** can hit the exact same target at the same speed. Which angle stays in the air longer? Why?

Written Explanation:

✏️ **Draw an Example Sketch:** (35° Flat vs. 55° High Arc)

Sketch both paths hitting fort:

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🤝 **Rival Partner Check:**
"I watched my partner take their shots and play fair!"
Partner Signature: _____

Mr. Mudry Stamp:
(Turn in at bell)

TEACHER STAMP / SCORE